Comparative Study and Practical Analysis of Generative AI Models

Autoencoder • VAE • DCGAN • Stable Diffusion

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CO3 + CO4

Four models, two jobs

AE

Autoencoder

Encoder squeezes the image into a latent code; decoder rebuilds it. Minimises reconstruction error.

VAE

Variational AE

Encodes a distribution (mean, variance) and samples from it. Loss adds a KL term.

GAN

DCGAN

Generator and discriminator compete. Convolutional layers make images from noise.

SD

Stable Diffusion

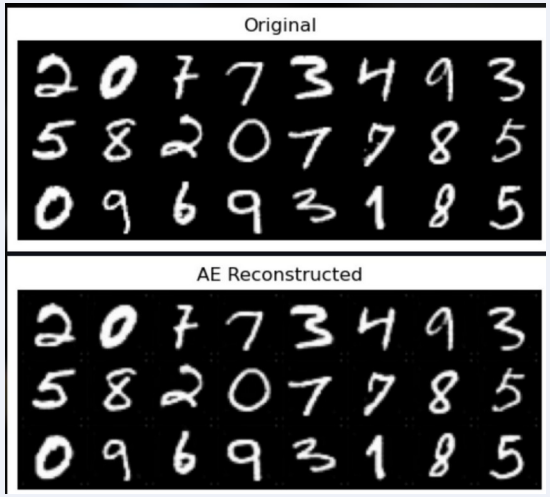
Pretrained model that denoises step by step in latent space, guided by a text prompt.

Comparison at a glance

	AE	VAE	DCGAN	Stable Diffusion
Core idea	Compress and rebuild	Latent distribution + KL loss	Generator vs discriminator	Iterative denoising with text guidance
Strength	Simple, fast, accurate reconstruction	Smooth latent space, can generate	Sharp, fast one-pass generation	Most realistic, prompt control
Limitation	Poor at creating new images	Blurry outputs	Unstable, mode collapse	Slow, needs GPU, odd hands or text
Output	Close to input, slight blur	Smoother, blurrier than AE	Sharp but small, artefacts	Detailed and photorealistic

Applications: AE denoising and compression | VAE anomaly detection | DCGAN augmentation | SD text-to-image art

Practical results from Google Colab



AE reconstruction

Almost identical to the original digits; faint dots on the background.



DCGAN

Readable 28x28 digits, but many are broken or ambiguous.



Stable Diffusion

Sharp, detailed and realistic; slightly over-saturated look.

Conclusion and reflection

Key learning

- AE gives accurate reconstruction; VAE gives a smooth latent space
- DCGAN is fast but unstable
- Diffusion gives the best realism at higher cost
- Pick the model by need: fidelity, speed or quality

AI usage and verification

- Tools: Claude for drafting and explanation
- Prompts: comparison table, VAE vs AE blur
- Checked against original papers and PyTorch / Hugging Face docs
- Results verified from our own Colab outputs